

Experiment 31

Aim

To perform morphological opening on an image using OpenCV.

Procedure

1. Import the required libraries, OpenCV and NumPy.
2. Read the input image.
3. Convert the image into grayscale.
4. Create a morphological kernel.
5. Apply the Opening operation using cv2.morphologyEx().
6. Display the original and opened images.
7. Save the resulting image.

Code:

```
import cv2
import numpy as np
img = cv2.imread("image.jpg")
if img is None:
    print("Error: Image not found!")
    exit()
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
kernel = np.ones((5, 5), np.uint8)
opening = cv2.morphologyEx(gray, cv2.MORPH_OPEN, kernel)
cv2.imshow("Original Image", img)
cv2.imshow("Opening Image", opening)
cv2.imwrite("opening_image.jpg", opening)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Sample Input:



Output:



Result

The morphological opening operation was successfully performed using OpenCV. The opening operation removed small unwanted objects and noise from the image.